

Response to the Editor and Reviewers

Manuscript MET-2026-0505

A Tutorial in Dynamic Systems Simulation and Modelling With R, ctsem, and lme4: Couples' Affect Dynamics Over Time

Dear Editor,

Thank you for the opportunity to submit a revised version of our manuscript to *Psychological Methods*. We appreciate the revision decision and the reviewers' detailed comments. We have organized the revision around the issues emphasized in the editorial decision: focus and length, scope and boundary conditions, and generalizability beyond the running example.

The revised manuscript now states the methodological contribution more explicitly in *[Introduction]* as elements of a workflow: theory is translated into a generative model, the model is simulated using understandable R functions, the corresponding statistical model is estimated, model-implied behavior is checked against the intended theory and observed data, and the specification is then revised, simplified, or extended. This workflow is not specific to *ctsem*, although *ctsem* provides a useful implementation for the continuous-time state-space models considered.

We have therefore revised the manuscript by making the conceptual workflow, scope conditions, model checking, and generalizability more explicit rather than simply adding caveats. We added *[Boundary Conditions]* and *[Scope and Outline]* near the outset, strengthened the *[Introduction]* and *[Conclusion]*, clarified the transition from simulation to state-space estimation in *[Univariate Dynamic Systems in Continuous Time]*, moved *[Model Evaluation and Comparison]* (including *[Practical Troubleshooting]*) earlier in the sequence, and added broader psychological applications when new concepts are introduced.

For ease of reviewing, we have included an html file showing the difference between the original and revised manuscript.

Editor

Editor Comment 1. *Both reviewers had concerns about the paper length. I see the same. I think the paper attempts to cover too much ground; a more focused narrative would strengthen the impact. One possibility is to focus on the ctsem package, and move the rest into the appendix.*

Response. We agree that the manuscript covers a wide range of material, and have revised to provide a more focused and structured narrative, as well as condensed or removed material that was repetitive or overly focused on code, plots, or local explanation rather than the main conceptual progression. We believe retaining the initial *lme4* examples is useful for didactic purposes, and hope the added clarity and structure is sufficient to address the concern.

We appreciate the paper, while reduced in length, is still long. We believe such a tutorial is a valuable resource for researchers interested in dynamic systems modelling, and hope the emphasis on structure and navigation goes

some way to mitigating the concern about length. *[Scope and Outline]* now includes explicit reader guidance, for example:

“Readers less familiar with dynamic systems modelling may wish to focus on Modelling Univariate Growth, Univariate Dynamic Systems in Continuous Time, and Model Evaluation and Comparison, leaving other sections for a later pass.”

If need be a further revision could drop *[Time-Varying Parameters]* or *[Interventions and External Events]*, but they are easy to skip now and the concepts they demonstrate allow for a much more general approach to dynamic systems modelling, making the tutorial much more valuable for more advanced readers.

Editor Comment 2. *Also suggested by the two reviewers, the manuscript should include a dedicated discussion on the scope and boundary conditions of the proposed models. Specifically, the authors should address the ‘big picture’ requirements and limitations of the approach—detailing the necessary data density for complex models, identifying specific scenarios where simpler models are preferable, and acknowledging practical failure points or empirical problems that the current tutorial does not cover.*

Response. We agree and have added *[Boundary Conditions]* near the outset. The tutorial now states more clearly when continuous-time models are appropriate and when simpler alternatives may be preferable:

“They are not automatically preferable, however. Simpler descriptive, mixed-effects, or discrete-time models may be better choices when the research question is not dynamic, when the time scale is not theoretically meaningful, when measurements are too sparse or unreliable to identify the proposed dynamics, or when the parameters of a more complex model would not be substantively interpretable.”

We also clarified in the same section that model complexity should be calibrated in both directions—models can be too complex for the data, but also too simple when distinct influences are bundled into one parameter. The discussion also notes data density, measurement quality, irregular timing, missingness, nonstationarity, floor or ceiling effects, non-Gaussian outcomes, and common estimation failures.

Editor Comment 3. *The tutorial relies on a simplified example concerning couples’ affect dynamics. While the example is effective pedagogically, discussions need to be given how the methodology and R package can be useful in other settings. The boundaries need to be defined (related to the comment above).*

Response. We agree that the manuscript should do more to make the generality of the workflow explicit. We retained the single running example because it allows the reader to see how the same substantive system can be developed from simple to complex specifications, making the consequences of each new model component easier to understand. However, discussion

points and explanations are made more general throughout the manuscript, and alternative examples from different domains are discussed when new concepts are introduced.

Editor Comment 4. *If you have not already done so as part of your Author Note, please provide the details (2-4 sentences) of prior dissemination of the ideas and data appearing in the manuscript (e.g., if some or all of the data and ideas in the manuscript were presented at a conference or meeting posted on a listserv, shared on a website, etc.).*

Response. We have added the requested prior-dissemination statement to the Author Note, noting the PsyArXiv preprint at https://osf.io/preprints/psyarxiv/4q9ex_v3 and the accompanying OSF project at <https://osf.io/xh6ue/>.

Reviewer 1

This tutorial paper explains how to build a dynamic model and how to estimate versions of such model with ctsem package. I think we need a lot more thoughtful iterative model building and more accessible resources on this topic. This tutorial both shows how to do that for a simple example and also showcases the capabilities of ctsem, both of which are useful. I have no technical concerns. However, while reading the paper I tried to collect some didactic points which might help the authors to further strengthen their tutorial so that it is more helpful for their target audience. I have a couple of larger points, and then a lot of smaller points below.

We thank Reviewer 1 (arbitrary number assigned for ease of responding) for the careful and constructive review. We found the comments highly useful for improving the didactic structure of the tutorial, especially the transition into continuous-time state-space modelling, the motivation for simulation, and the practical interpretation of model components.

Major Points

Reviewer 1, Major Point 1. *Explain general rationale more at the outset: The introduction says that you will walk the reader through iterative model building. However, it is perhaps not obvious to everyone that this means that you will create generative models based on theory and then recover these models from synthetic data. It is not really explained why this is happening. I think this would be good to lay out in some more detail, so that the reader can better appreciate the material that follows. Now this is left implicit, and a reader with little background in modeling might be wondering why you are not just fitting successively more complex models.*

Response. The revised *[Introduction]* now explains why the tutorial begins from generative models and simulation rather than from fitting successively more complex models to data alone. It states the workflow explicitly:

“The central contribution is the components of a workflow: translate theory into a generative model, simulate that model with transparent R functions, estimate a corresponding statistical model, check whether the model-implied behavior matches the intended theory and observed data, and then revise, simplify, or extend the specification.”

It then justifies simulation as the starting step:

“Simulation is a core element in a modelling workflow because it can make the consequences of theoretical assumptions visible before empirical estimation. It is of course important to recognise that simulated data will not resemble every empirical complication, but this approach helps understand how model parameters generate trajectories, when different sources of variability have different implications, and how a fitted model can fail even when it explains a large amount of variance.”

Reviewer 1, Major Point 2.

Jump in difficulty when introducing ctsem: I think there is a big jump in the difficulty of the material when transitioning to the section on ctsem. Before things are still quite easy to follow and you still provide all the code. But on page 11 the difficulty greatly increases. New ideas are introduced much more quickly, and I don't feel anymore taken by the hand and guided through everything like in the first 10 pages of the manuscript. For example, you just mention the latent dynamic model in passing, without ever really motivating or discussing it in the context of the running example. Figure 18 is also not really explained in great detail, which might be difficult to process especially for the less technically inclined readers, who I suspect are your target audience. This is handled better in Figure 14 though, where there is more of a walk-through. In any case, I felt like the difficulty level was increasing quite a bit. This might be addressed by explaining a bit more here and there. But also perhaps by warning the reader that this will happen.

Response. We revised *[Univariate Dynamic Systems in Continuous Time]* to provide a clearer bridge from simulation to state-space estimation in `ctsem`. In *[State-Space Models]*, we now motivate the latent dynamic model for the running example:

“Instead of treating the observed affect score as the process itself, we distinguish a latent dynamic process from the observations that measure it. The latent process evolves contin-

uously according to the dynamic model, while the measurement model links that latent process to the observed affect scores at the times they were measured.”

Reviewer 1, Major Point 3.

Paper Length: The paper is pretty long and you have 32 (!) figures in the main text. As someone who is reasonably knowledgeable of what you are presented here and who found the paper a very interesting read, I found it challenging to read it in one sitting. I could imagine that it would take readers with less of a technical background even more resolve. This can be fine if you see the paper more of a handbook where one can look things up. But if the idea is that people read the whole paper front to back, you could consider shortening the material a bit. One option could be to only sketch the later models in the main text and move the details into the appendix.

Response. We have focused on making the tutorial easier to navigate, including stronger sectioning, a more explicit roadmap in *[Scope and Outline]*, and guidance on which sections suit different readers (see our response to Editor Comment 1). We also condensed repetitive code-heavy passages, plots, and explanations where they did not add substantially to the conceptual progression.

Reviewer 1, Major Point 4.

Possible Models and Behavior: I was wondering whether you could comment a bit more in the end on the possibility of nonlinear models. For example, can the latent processes interact? That of course would allow one to model systems with much more complex behaviors than the ones in the running example, which I think were all single fixed point systems. Related, estimating initial states and using the (V)AR(1) transient to get a nonlinear effect towards the single fixed point is a great use of this model class. However, this of course does not provide a model for how a person ended up in their initial state in the first place. This is perhaps another higher-level topic to comment on in the end.

Response. We replaced one time-dependent-moderator example with latent state moderation in *[Latent State Moderation of System Parameters]* (*[Time-Varying Parameters]*), as it is a more general approach and useful for the reader to see. The emphasis of the paper remains on developing the understanding with linear models; we do not dive into elaborate nonlinear theory. *[Further Works and Resources]* briefly notes additional nonlinear extensions.

Reviewer 1, Major Point 5.

Data Requirements: Since the paper opens very generally with dynamical systems, perhaps you could also comment on the data requirements of ctsem. Clearly the

complexity of models must be limited by the amount of data that is available.

In addition, it would be interesting to hear the author's big picture view on creating dynamical systems models in psychology. Specifically, do you think that we can always fit the whole model to a single dataset like you do here? If not, what are alternative approaches? How could one in principle combine different sources of information to recover a dynamical system? Of course, I don't expect you to solve any of these issues, but a (brief) discussion of these questions could help relate your approach to other approaches.

Response. We added discussion in *[Boundary Conditions]* on when models are too complex or too simple for the data (see also our response to Editor Comment 2). *[Introduction]* also notes that simulation can help researchers gauge approximate data requirements under specific assumptions, while clarifying that this is not a power-analysis tutorial. Combining different sources of information for systems estimation is a fascinating topic but beyond the scope of this tutorial.

Minor Points

Reviewer 1, Minor Point 1.

Small thing: On page 2 in the third list point you conceptualize noise as "reflecting unpredictable events", which almost makes this sound like a limitation. But I guess these are just the causal effects on the modeled variables that we choose to abstract away with a probability distribution?

Response. We removed the "reflecting unpredictable events" wording from *[Scope and Outline]* and reframed randomness in *[State-Dependent Change with Fluctuations]* as a deliberate abstraction of unmodelled influences rather than a limitation of the model:

"In our model so far, affect depends only on its current level and a fixed intercept, but in reality many other variables also matter: sleep, minor hassles, chance encounters, and countless other influences we do not measure or include explicitly. System noise stands in for the combined effect of these unmodeled variables: it allows the true process state to be pushed up or down by influences outside the equation."

We also clarify how this process noise differs from measurement error:

"Unlike measurement error, which affects only the observation, system noise changes the underlying affect level itself,

and those shifts persist because the dynamics carry them forward over time.”

Reviewer 1, Minor Point 2.

Page 2: “our first idea”; I don’t follow why this would be the simplest idea you can come up with. I would think the simplest idea is that nothing changes.

Response. We revised the opening of *[Linear Change]* so the first example is framed as a minimally dynamic linear-change model rather than the simplest conceivable case.

Reviewer 1, Minor Point 3.

page 3: I found it very curious that you conceptualize $\epsilon(t)$ as measurement error. Is this really the most interesting stochasticity to model here?

Response. We clarified in *[Linear Change]* that measurement-error or unsystematic noise is introduced first; *[State-Dependent Change with Fluctuations]* then introduces system noise as stochastic input propagated through the dynamic system.

Reviewer 1, Minor Point 4.

page 3: I was missing a comment on the (measurement) scale of affect, and the choices of the parameters (trend, noise variance). Of course, I understand that you want to keep this concise, but if you go for this first-principle tutorial (which I think is great), it feels odd to just introduce these unmotivated parameter values.

Response. We added a brief explanation in *[Linear Change]*:

“We use an arbitrary affect scale where larger values indicate more positive affect; the starting level, slope, and noise values are chosen to make the implied trajectory visible rather than to represent calibrated empirical values.”

Reviewer 1, Minor Point 5.

Page 4: you show the correlation between random effects to be -0.92. It is not entirely clear from the text before why we would expect that.

Response. We revised *[Individual Differences in Slopes]* to make the intercept–slope correlation construction more explicit in the simulation code and text.

Reviewer 1, Minor Point 6.

Page 4: in the model with 20 subjects, the measurement error variance is 0.1, while above for the single person it is 1. Why?

Response. We harmonised the measurement-error standard deviation to 0.2 throughout the manuscript.

Reviewer 1, Minor Point 7.

Figure 2: Maybe you can color the people along their intercept values, to make this correlation between the random effects visually more obvious.

Response. We implemented this in *[Individual Differences in Slopes]* by colouring trajectories by initial affect level.

Reviewer 1, Minor Point 8.

On page 6 you say "The solid line does not map well onto the observed trajectory of affect, and there is a large amount of uncertainty about the model-implied trajectory." perhaps you can also briefly comment on this "mismatch" (regularization due to the empirical bayes of the multilevel model). But I'm not sure anymore what is happening here (see next comment).

Response. We added a short explanation in *[Individual Differences in Slopes]*:

"This is related to partial pooling: before conditioning on the subject's own observations, the model relies on the population distribution of trajectories rather than on the specific intercept and slope for this person."

Reviewer 1, Minor Point 9.

Page 6: I did not understand what is shown in Figure 4, which I think you refer to with "the second plot".

Response. We revised the figure caption and text references throughout so that the displayed quantities are identified directly rather than by ambiguous positional descriptions.

Reviewer 1, Minor Point 10.

Page 6: You write "By formalizing our linear growth process for Jill and other patients, we observe that our simple conceptual idea leads to a surprising prediction. Namely, therapy actually helps people with a lower initial level of affect to have a higher affect level at the end of the 21 weeks of therapy, compared to people who started with higher affect." I find it a bit odd to frame this as a discovery of the analysis, because this is the model we specified and simulated from.

Response. We revised the wording in *[State-Dependent Change (Discrete Time)]* so implications are framed as following from the assumed model:

"By formalizing our linear growth process for Alex and other patients, we can inspect what our assumptions imply."

Reviewer 1, Minor Point 11.

On pages 6-7 you introduce a non-linear effect via the transient of an AR(1) process. I was a bit confused by this, because before you modeled the linear effect with a deterministic linear trend. So for me the logical step would have been to model this with a non-linear deterministic trend. Not saying that this is what you should do in practice, but that would have followed more logically. Especially, since you do not announce this conceptual switch. I think it would be good to discuss this change in the modeling approach more explicitly.

Perhaps also comment on the problem that this is perhaps a bit of an odd model for therapeutic change, because when you start above B , your affect will decrease.

Response. We revised [*State-Dependent Change (Discrete Time)*] to announce the conceptual shift explicitly:

“This is a conceptual shift from a deterministic trend indexed only by time to a state-dependent process: change depends on the current level of affect, or change is ‘state-dependent’. It also has substantive implications. If a person’s affect starts above the implied long-run level, the same model predicts movement downward rather than continued improvement.”

Reviewer 1, Minor Point 12.

In the first paragraph on page 8 you mention some advantages of CT over DT modeling. If you won’t expand on this later, I think it would be good to do this somewhere. Currently, you can only appreciate these points if you already know about them.

After reading a bit more, I feel like it would be good to better motivate CT already here. Because you basically have to give a mini-calculus course here, and especially people who don’t find this easy might not understand why they have to go through all this trouble, if they didn’t really understand why CT is tangibly better than DT.

Response. We strengthened the motivation for continuous-time modelling in [*Introduction*] and at the opening of [*Univariate Dynamic Systems in Continuous Time*]:

“Continuous-time representations are particularly valuable for theory-oriented model development, because of the more natural link between continuous-time parameters and typical theoretical interests. Because many psychological processes — such as emotion regulation, stress responses, symptom fluctuation, learning, physiological regulation, or cognitive fatigue — are thought to evolve continuously rather than only when measured, continuous-time models provide a natural representation. Parameters such as drift (state dependence), diffusion (system noise), and external inputs align much more closely with mechanistic interpretations, instead of as artifacts of the measurement schedule as with discrete-time models. Continuous-time models are also useful when observation intervals vary or when the time scale of change is substantively important.”

Reviewer 1, Minor Point 13.

Making the step from discrete time to continuous time could be softened by rewriting the DT model into a difference equation, and then say that the time step becomes arbitrarily small, giving us the derivative on the LHS.

Response. We added this bridge in [*Differential Equations*], starting from average change over a finite interval before introducing the derivative:

“One way to soften this shift is to start with average change over a finite interval. As the interval Δt becomes smaller, this average rate approaches the instantaneous rate of change,

written as the derivative $\frac{dn}{dt}$. The differential equation then describes what determines that momentary rate of change.”

Reviewer 1, Minor Point 14.

Linear ODE: Perhaps you can show the function eta and then take the derivative to get B. Maybe this clicks better for people who ever had calculus and remember the power rule.

Response. We added a short calculus-based explanation in [Differential Equations] linking a linear trajectory to its constant rate of change.

Reviewer 1, Minor Point 15.

Page 8: When switching to the simulation, now the start of "Simulation" is very technical. Perhaps you could explain a bit more which problem you have to solve here and how you solve it with this numerical approximation.

Response. We revised the opening of [Simulating Continuous-Time Dynamics] to explain why numerical approximation is needed before presenting the Euler method.

Reviewer 1, Minor Point 16.

Page 8: Euler's method: I was surprised to see that you only take 20 time steps. Don't you usually need more to get a good approximation of the analytical solution (which here you could still derive)? It might also be confusing for the reader, because basically you are doing the same as in the DT case.

Okay, I now realize that you discuss this right after. Perhaps add a sentence saying that you will first approximate this and then make this more precise.

This could also be a place to cut material and make the paper shorter. That is, immediately add many steps.

Response. We clarified in [Simulating Continuous-Time Dynamics] that the initial Euler approximation is intentionally crude for didactic purposes, then refined the step size to improve the numerical solution.

Reviewer 1, Minor Point 17.

Page 8: You here for the first time introduce stochasticity that is pushed through the system and that is not measurement error. I think it could be nice to discuss more what we are modeling with this, because I think conceptually this is a big step.

Response. We expanded the discussion in [State-Dependent Change with Fluctuations].

Reviewer 1, Minor Point 18.

infinitesimal: I'm not a mathematician myself, but my understanding is that this has been replaced with the limit definition / and "arbitrarily small" (and not: infinitely small, which is not a real number).

Response. We revised this language in *[Simulating Continuous-Time Dynamics]* to use “arbitrarily small” and limit-based wording where appropriate.

Reviewer 1, Minor Point 19. *I’m not sure whether “system noise” was properly defined at some point.*

Response. We now define system noise at first use in *[State-Dependent Change with Fluctuations]* and distinguish it from measurement error.

Reviewer 1, Minor Point 20. *At the end of page 10, when introducing ctsem you mention “state-dependence” but it’s not clear to me how this relates to the above. In general, the switch is pretty abrupt, because you introduce ctsem as a way to fit a latent dynamical model. But so far, we did not look at such models, so this came a bit out of nowhere for me. Reading further on, I think it would be good not only to introduce the state-space latent-measurement model setup, but also motivate why this is useful in the case of the running example.*

Response. We agree and revised the transition in *[State-Space Models]*.

Reviewer 1, Minor Point 21. *Figure 8: Perhaps it would be good to briefly talk the reader through the model structure again. This could also tie the earlier discussion of all pieces together.*

Response. We removed the initial univariate matrix model-structure figure in *[ctsem: A Flexible Tool for Dynamic Systems Modelling]* as it partially duplicated the later walk-through in *[Modelling Multivariate Systems]*, where the matrix view is more critical.

Reviewer 1, Minor Point 22. *Figure 10 and 11: Similarly to above, I am not sure why we are looking at this analysis.*

Response. We revised the text to state the purpose of these figures more explicitly: they show how the fitted continuous-time model implies trajectories and uncertainty at observed and unobserved time points, help to provide intuition for how the model works, and also provide useful checks on the model, in case predictions diverge substantially.

Reviewer 1, Minor Point 23. *Page 14: Jill continues “Consider Jill, who attends weekly therapy sessions during her 20-week treatment period. Each session provides a temporary boost to her affect. The longer term effect of the therapy is already accounted for by the growth trajectory we have already specified in earlier examples.” This is where the running example for me finally departs from reality. I don’t think that this is a fair representation of how any therapy works. Not sure*

how to deal with this. Perhaps just acknowledge it? I don't know.

Response. In *[Impulse Effects]* we expanded slightly on this to at least make clear the perhaps implausible assumptions.

Reviewer 1, Minor Point 24. *Perhaps you can choose two neutral names to make the couple example less hetero-normative. Like Alex and Robin or something like that.*

Response. We revised the running example throughout to use the neutral names Alex and Robin.

Reviewer 1, Minor Point 25. *page 16: "Correlation in system noise, random life events that cause fluctuations in Jill's affect trajectory are also likely to contemporaneously impact Bert's affect (and vice versa)." I think it would be interesting for the reader to have spelled out why this is the case.*

Response. We added examples in *[Modelling Multivariate Systems]*:

"Such shared influences could include a household stressor, a conflict, a shared positive event, or another unobserved environmental input that affects both partners."

Reviewer 1, Minor Point 26. *page 23-24: Adding time-dependent moderators: you give the dummy-coded variable indicator communication as an example. Can you also add interaction terms between the modeled state-variables? So, in your example, the cross-lagged effect of Jill's affect on Bert's affect depends also on Bert's current affect.*

Response. Instead of two examples with time dependent moderators, we replaced one with an example that demonstrates moderation based on a latent process instead, so readers are more aware of such possibilities in *[Latent State Moderation of System Parameters]*.

Reviewer 1, Minor Point 27. *In the model selection you seem to use within-sample R^2 . However, wouldn't this R^2 only go up when estimating models that are too complex? When they are nested, they would technically not be misspecified, but of course you would never be able to select a data generating simpler/constrained model.*

Response. We revised *[Model Evaluation and Comparison]* to avoid implying that within-sample R^2 is sufficient for model selection:

"A more flexible model will often improve within-sample summaries, but this does not mean the extra parameters are interpretable or supported. Conversely, a model that is too simple may leave distinct sources of variation bundled into one

parameter. The examples below therefore use R^2 -style summaries only as one descriptive starting point, and then focus on prediction plots, residual autocorrelation, and posterior-predictive-style checks. This should not be treated as a stand-alone model selection criterion: complex models can improve within-sample explained variance while still implying the wrong process.”

Reviewer 2

This manuscript presents a tutorial on dynamic systems simulation and modeling using R, ctsem, and lme4, organized around a running example of couples’ affect dynamics over time. It is ambitious in scope, pedagogically oriented, and clearly intended to help readers move from familiar mixed-effects models toward more complex dynamic systems formulations. The emphasis on simulation, visualization, and iterative model-building is potentially valuable. My central concern is that the paper remains substantially closer to an extended software tutorial than to a strong methodological article. It demonstrates many modeling possibilities, but it does not yet distill them into a sufficiently clear, general, and high-level methodological contribution. Overall, I have some major concerns about this paper.

We thank Reviewer 2 (arbitrary number assigned for ease of responding) for the detailed critique. Several comments identified places where the manuscript needed clearer statements of contribution, generalizability, assumptions, and practical limitations. We agree that these aspects were not sufficiently explicit in the original version. We retained the worked tutorial structure, however, because it is central to the manuscript’s purpose: *Psychological Methods* publishes tutorials, and the paper is intended to build intuition and understanding for dynamic systems modelling by working through detailed examples. Indeed there is significant ctsem specific content, but we hope the refinements make the general methodological contribution clearer.

Reviewer 2, Main Concern 1.

The manuscript does not yet establish a sufficiently strong methodological contribution beyond package documentation. At present, the paper reads primarily as a long implementation-oriented tutorial. It walks the reader through code, model setup, and progressively more complex examples, but it does not clearly demonstrate what the paper contributes methodologically beyond what could be learned from package vignettes, workshop materials, or a well-developed software tutorial. The authors need to extract and explicitly articulate a small set of transferable principles, rather than relying on the accumulation of examples to imply the contribution.

Response. We believe that a well-developed tutorial relying on openly available tools aligns with the aim of *Psychological Methods* to disseminate methods for analyzing, understanding,

and interpreting psychological data. Nevertheless we appreciate the suggestion to more clearly explicate the methodological contribution. The revised *[Introduction]* states:

“The central contribution is the components of a workflow: translate theory into a generative model, simulate that model with transparent R functions, estimate a corresponding statistical model, check whether the model-implied behavior matches the intended theory and observed data, and then revise, simplify, or extend the specification.”

It also emphasises that the principles demonstrated are not specific to `ctsem`:

“While this tutorial emphasises the `ctsem` package, and specific syntax and package issues are discussed, the principles demonstrated are general and can be applied to other packages and modelling approaches, as well as to discrete-time forms (also available in `ctsem` but not emphasised here).”

We retained the worked examples because they are the mechanism through which this workflow is taught. Although `ctsem` is used for estimation and visualization, the simulations themselves are coded with basic R functions, the state-space model structure is quite general, and the various modelling considerations and approaches are not specific to `ctsem`. The package therefore serves as one implementation route for a broader set of principles: mapping theory to model structure, separating measurement from underlying process, reasoning about time scale and interacting variables, and checking whether fitted models behave as intended.

Reviewer 2, Main Concern 2.

The paper’s generalizability is limited by its heavy reliance on a single didactic example. The running example of couples’ affect dynamics is pedagogically effective, but the manuscript remains too tightly tied to this particular scenario. As a result, the paper sometimes feels less like a broad methodological tutorial and more like a long, well-developed demonstration within one narrow substantive context.

Response. We retained the running example because it allows the tutorial to show how model complexity develops cumulatively from simple to complex specifications, but have made explanations and examples more general throughout, and included alternative examples from different domains whenever new concepts are introduced, to address the concern.

Reviewer 2, Main Concern 3.

The tutorial is built largely around hypothetical or simulated data-generating mechanisms. This is understandable for pedagogical purposes, but it also limits

the manuscript’s practical credibility. Real psychological data often involve irregular spacing, substantial missingness, nonstationarity, limited measurement precision, floor/ceiling effects, non-Gaussian variables, and model misspecification that is more ambiguous than the examples shown here. The manuscript needs a dedicated section on scope conditions, empirical limitations, and practical failure points. At minimum, the authors should discuss: when these models are likely to be appropriate, when simpler models may be preferable, what kinds of data are insufficient for the more complex models, and what common empirical problems are not addressed by the tutorial.

Response. We agree that the manuscript should be clearer about these aspects. We use simulated data deliberately because understanding models is fundamental to responsible modelling, and generating data from known assumptions is one of the clearest ways to learn what those assumptions imply; *[Introduction]* now states this explicitly:

“It is of course important to recognise that simulated data will not resemble every empirical complication, but this approach helps understand how model parameters generate trajectories, when different sources of variability have different implications, and how a fitted model can fail even when it explains a large amount of variance.”

We have made model checking more central to the manuscript by moving *[Model Evaluation and Comparison]* earlier in the tutorial sequence and expanding on its importance and practical considerations. We have also added an explicit discussion of scope conditions and practical failure points in *[Boundary Conditions]*:

“They are not automatically preferable, however. Simpler descriptive, mixed-effects, or discrete-time models may be better choices when the research question is not dynamic, when the time scale is not theoretically meaningful, when measurements are too sparse or unreliable to identify the proposed dynamics, or when the parameters of a more complex model would not be substantively interpretable.”

The revised manuscript emphasises that real data likely involve varieties of complexity that are not fully captured by the simplified examples. The same section also clarifies that model complexity should be calibrated in both directions:

“A model can be too complex for the data, leading to weak identification, unstable estimates, or estimation failures. A model can also be too simple however: distinct influences

may be bundled into a single parameter and then interpreted as if that parameter represented one psychological process. Important empirical complications not fully treated here include missing-not-at-random, nonstationarity, floor or ceiling effects, non-Gaussian outcomes, and many other forms of misspecification.”

We added a practical troubleshooting discussion in *[Practical Troubleshooting]* covering common estimation problems and how readers should respond when models fail.

Reviewer 2, Main Concern 4.

The manuscript repeatedly frames the approach as continuous-time SEM and state-space modeling. However, much of the tutorial relies on highly simplified single-indicator setups. That may be acceptable pedagogically, but the paper does not adequately acknowledge the consequences of that simplification. As written, the paper often reads more like a tutorial on continuous-time dynamic regression with measurement error than a tutorial that fully leverages the latent-variable advantages of continuous-time SEM.

Response. The revised *[State-Space Models]* section now states:

“The examples in this tutorial use simple single-indicator measurement models for pedagogical clarity. Thus, the tutorial emphasizes the dynamic state-space part of continuous-time structural equation modelling more than full multiple-indicator latent-variable measurement modelling.”

We also note that this focus reflects a broader trend in psychological methods toward modelling multivariate systems, networks, and coupled processes, where the key substantive questions often concern dynamic relations among states rather than only latent factors explaining many observed indicators.

Reviewer 2, Main Concern 5.

The manuscript needs stronger guidance on assumptions, identification, and model choice. The tutorial is effective at demonstrating how one can move from simple to more complex models. However, it is far less effective at telling readers when they should do so, and on what basis. This is a serious omission. A strong methods tutorial should not only demonstrate that a model can be fit; it should also help the reader decide: when linear growth is inadequate, when state dependence is warranted, when system noise should be distinguished from measurement error, when interactions or time-varying moderation are theoretically justified, and when data are too weak to support these expansions.

Response. We revised the manuscript to make the rationale for model extensions more explicit, and placed an expanded model

evaluation section earlier in the tutorial sequence. We frame model choice as triangulation among theory, measurement design, observation density, model-implied behavior, diagnostics, and parsimony, rather than as a clean automatic selection rule.

Reviewer 2, Main Concern 6.

The later sections on misspecification and posterior predictive checks are among the better parts of the manuscript. However, the paper still does not provide enough practical guidance on the difficulties of estimation. The authors should add a practical troubleshooting section that explicitly discusses common failure modes and how readers should respond to them. The paper should do more to teach caution, not only implementation.

Response. We added *[Practical Troubleshooting]* within *[Model Evaluation and Comparison]*:

“Dynamic state-space models can fail in ways that are informative. Common warning signs include Hessian warnings after fitting or extremely large or small standard errors / uncertainty intervals, unstable estimates across small changes in specification, parameter estimates at boundary values, implausibly fast or slow dynamics, and poor posterior predictive behavior.”

Reviewer 2, Main Concern 7.

The authors note in the cover letter that the manuscript is longer than typical submissions because of figures and code. I agree that the paper is currently overlong for its central contribution. The issue is not merely page count; it is also information allocation. Too much of the main text is occupied by details that could be moved to supplementary material without reducing the paper’s conceptual value.

Response. We cautiously acknowledge this concern. but emphasise that the article is intended as a tutorial that walks through the details of a model-building workflow – we are not aiming only for conceptual value but very practical support. We have made targeted revisions rather than removing core tutorial detail: clearer structure and reader guidance in *[Scope and Outline]*, a more explicit workflow in *[Introduction]*, and condensation of repetitive code- and plot-heavy passages (see our response to Editor Comment 1).

Reviewer 2, Main Concern 8.

The conclusion is notably underdeveloped. It does not adequately distill what the reader should have learned, nor does it clearly summarize the limits of the proposed workflow.

Response. We substantially revised *[Conclusion]*:

“The central lesson of this tutorial is therefore not simply how to fit a particular software model, but how to move between theory, generative assumptions, simulation, estimation, model checking, and revision.”

It also summarises the limits of the workflow:

“These ideas are useful only to the extent that the model, data, and theory support the interpretation. A complex model that is weakly identified can mislead, but so can an overly simple model that bundles distinct mechanisms into one parameter.”

The conclusion also reiterates the need for cautious, iterative model building and revision when the model fails.

Specific Comments

Reviewer 2, Specific Comment 1.

There are multiple language and copyediting problems that should be corrected before the paper can be seriously evaluated in final form. A few examples include: "introduce a approaches", "vector autogression", "do not be mislead", "relatively-poor".

Response. We corrected these and other copyediting issues throughout the manuscript.

Reviewer 2, Specific Comment 2.

Some code examples appear inconsistent or insufficiently cleaned. Because this is a tutorial, code accuracy and consistency matter especially strongly. I noticed at least one example in which parameter assignments appear contradictory (e.g., $B <- .5$ followed immediately by $B <- 2$ in the same continuous-time simulation block). Even if harmless in context, such issues undermine reader trust in the tutorial.

Response. We reviewed the code examples and removed contradictory assignments, including the duplicate continuous-intercept assignment in the *[Simulating Continuous-Time Dynamics]* block.

Closing

We again thank the editor and reviewers for their careful feedback. The revised manuscript aims to retain the strengths of the tutorial format while enhancing the structure and readability of the tutorial, making the methodological contribution, generalizability, scope conditions, and practical cautions substantially clearer.

Sincerely,

Charles Driver & Michael Aristodemou